

Comparison: Risk Parity, Markowitz, and Equally Weighted Practice 5

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Who's the most robust?

EQUALLY WEIGHTED

- Stable portfolio
- Performed well in the testing sample too, sometimes with improvements, but not as good as the RP.

RISK PARITY

Always improving, in both ways of building it.

- Best testing performance when talking about sharpe ratio.
- Significant improvement in the 50-50 sample case.

MARKOWITZ

Good start, bad end

- Markowitz's maximum sharpe ratio is competitive only in training sample, small changes in sample data have led into big worsening.

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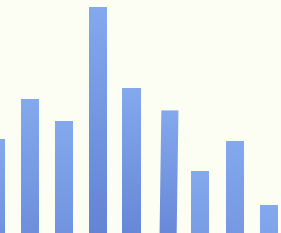
Case I

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Conclusion



(All results about
returns are expressed
as
daily-excess returns,
including costs)

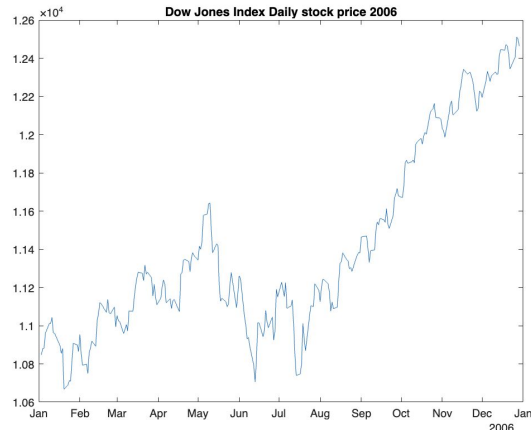
The background features a light green gradient with various financial-themed elements. In the top left, there is a candlestick chart with white and blue bars. A blue line graph with an upward-pointing arrow starts from the left and extends towards the top center. In the top right, there is a bar chart with blue bars of varying heights. Three blue coins with dollar signs (\$) are scattered around: one in the top right, one in the bottom left, and one in the middle right. The central text is enclosed within two concentric light gray circles.

01 Portfolio Composition

PORTFOLIO COMPOSITION

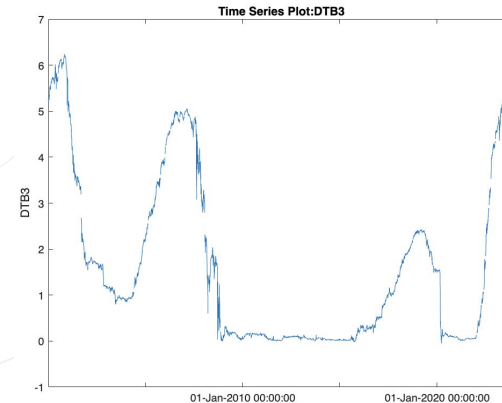
Dow Jones Index

Tracks the performance of 30 important publicly-traded US companies across different industries i.e technology or finance.



3-Month US Treasury Bill

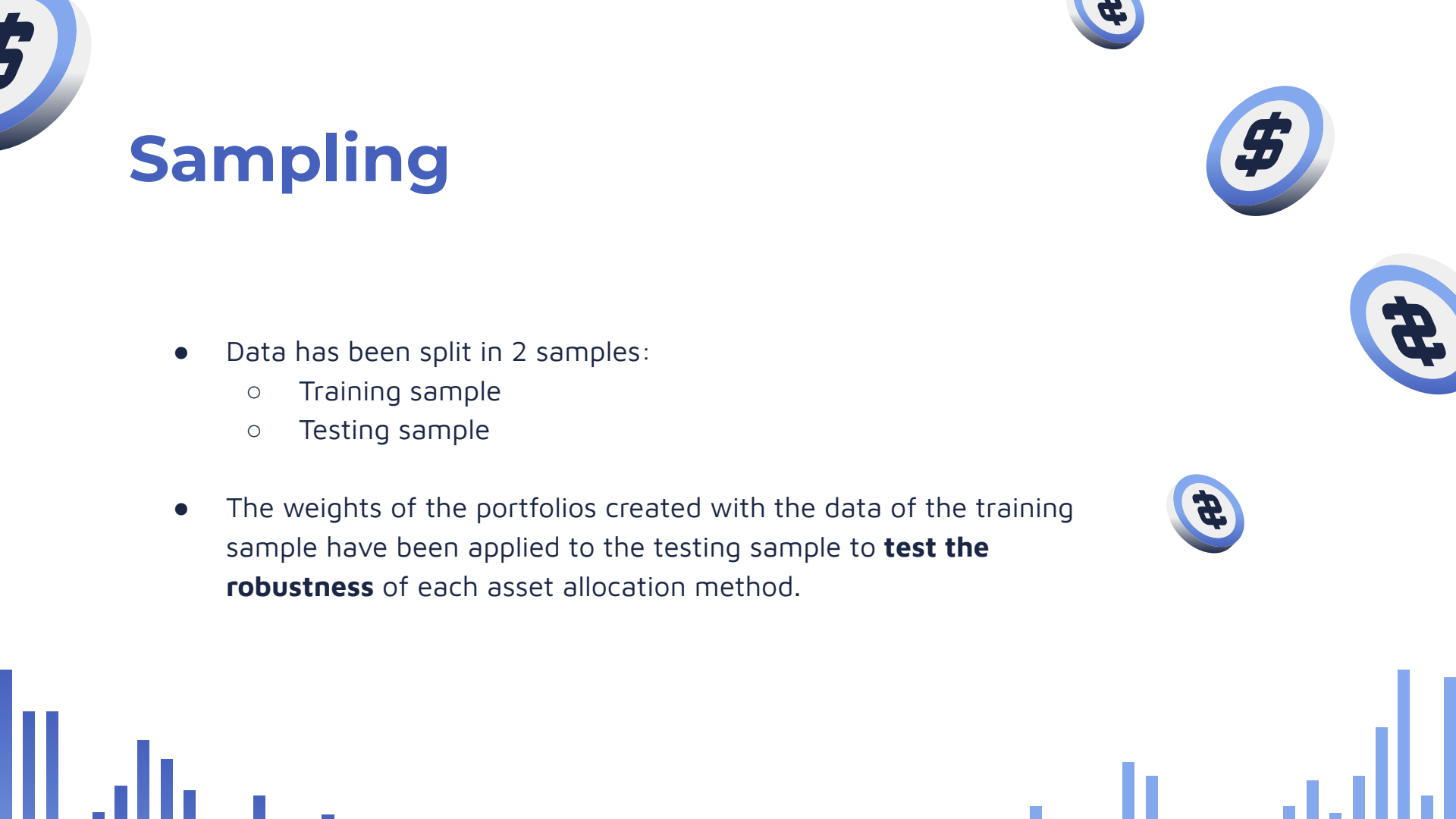
Cost of short-term borrowing for US government. Indicator of economic conditions and considered safe.



(Missing data have been filled in MATLAB)

TX - United Technologies Corporation
VZ - Verizon Communications Inc.
WMT - Walmart Inc.
XOM - Exxon Mobil Corporation





Sampling

- Data has been split in 2 samples:
 - Training sample
 - Testing sample
- The weights of the portfolios created with the data of the training sample have been applied to the testing sample to **test the robustness** of each asset allocation method.



Proportions

CASE 1

Training data $\leftarrow$ first 30% of the trading days in 2006

Testing data $\leftarrow$ last 70% of the trading days in 2006

CASE 2

Training data $\leftarrow$ first 50% of the trading days in 2006

Testing data $\leftarrow$ last 50% of the trading days in 2006

Asset allocation methods:

COST

0.5%

0.01%

0.6%

0.6%

Three portfolios have been created for the two samples in 2 different proportions:

- **Markowitz**: the portfolio with the maximum Sharpe Ratio among all the return-volatility combinations located on the efficient frontier.
- **Equally Weighted**: includes all 31 assets in the same proportion (1/31).
- **Risk Parity**: composed by 6 clusters, whose risk-contribution is equal. Each cluster contains assets with similar correlations, this allowed us to group them easily using the short formula (σ_i/σ_{tot}).
- **Simple Risk Parity**: aims to allocate investments such that each asset contributes equally to the overall portfolio risk. In this approach, the weight of each asset is set to be inversely proportional to its volatility (σ), where the volatility (σ) represents the standard deviation of the asset's returns.

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02

Sample 1



Markowitz I - Comparison

IN-SAMPLE

`riskOptimalM = 0.0086`
`retOptimalM = 0.0027`
`SharpeROptimalM = 0.3127`

OUT-OF-SAMPLE

`riskOptimalMO = 0.0117`
`retOptimalMO = -2.6477e-04`
`SharpeROptimalMO = -0.0226`



The data reveals that in this case the portfolio in the out-of-sample period heavily underperforms in comparison to the in-sample portfolio since

- Expected return decreases and becomes negative
 - Standard deviation or risk is greater: $0.0117 > 0.0086$
 - Sharpe Ratio suggests that the out-of sample portfolio is underperforming the risk-free asset and therefore losing money.
- 

Equal Weights I - Comparison

IN-SAMPLE

```
riskM_EW = 0.0060  
retM_EW = 5.8944e-04  
SharpeR_EW = 0.0987
```

OUT-OF-SAMPLE

```
riskM_EWO = 0.0065  
retM_EWO = 4.0660e-04  
SharpeR_EWO = 0.0626
```

The data reveals that in this case both portfolios behave similarly as:

- Expected returns are similar and very modest
- Standard deviations or risks remain close to each other, stable
- Sharpe Ratio for both portfolios remain positive but decreased in out-of-sample period.



Simple Risk Parity I - Comparison

IN-SAMPLE

```
risk_in_SimpleRP = 0.0057  
ret_in_SimpleRP = 1.5191e-04  
SharpeR_in_SimpleRP = 0.0268
```

OUT-OF-SAMPLE

```
risk_out_SimpleRP = 0.0062  
ret_out_SimpleRP = 3.6932e-04  
SharpeR_out_SimpleRP = 0.0599
```



The data reveals that in this case out-of-sample portfolio slightly outperforms the other due to:

- Expected return is slightly above the other
 - Standard deviations or risks is also higher
 - Sharpe Ratio shows that the risk-adjusted return for the second portfolio is also higher and thus, the latter is preferred.
- 



Risk Parity I - Comparison

IN-SAMPLE

```
risk_in_RP = 0.0061  
ret_in_RP = 1.3033e-04  
SharpeR_in_RP = 0.0214
```

OUT-OF-SAMPLE

```
risk_out_RP = 0.0081  
ret_out_RP = 6.0440e-04  
SharpeR_out_RP = 0.0746
```

The data reveals that in this case out-of-sample portfolio slightly outperforms the other as well since:

- Expected return is well above the other
 - Standard deviations or risks is also higher
 - Sharpe Ratio shows that the risk-adjusted return for the second portfolio is also higher and thus, the latter is preferred.
- 

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03

Sample 2



Markowitz II - Comparison

IN-SAMPLE

```
riskOptimalM = 0.0109  
retOptimalM = 0.0020  
SharpeOptimalM = 0.1795
```

OUT-OF-SAMPLE

```
riskOptimalMO = 0.0102  
retOptimalMO = -5.5988e-05  
SharpeOptimalMO = -0.0055
```



The data reveals that in this case the in-sample portfolio outperform the other because:

- Expected return is much greater than the other
 - Standard deviation or risk is though similar
 - Sharpe Ratio suggests that the out-of sample portfolio is losing money while the other is earning 0.1796 units of excess returns over the risk-free.
- 



Equal Weights II - Comparison

IN-SAMPLE

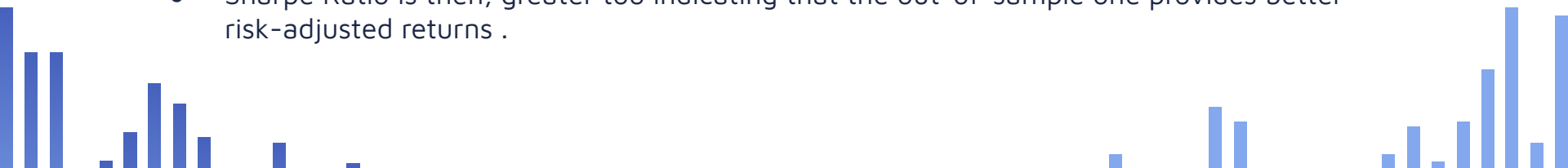
```
riskM_EW = 0.0070  
retM_EW = 3.5276e-04  
SharpeR_EW = 0.0507
```

OUT-OF-SAMPLE

```
riskM_EWO = 0.0056  
retM_EWO = 6.6349e-04  
SharpeR_EWO = 0.1176
```



The data reveals that in this case the out-of-sample portfolio excels in performance when comparing to the first as:

- Expected returns are doubling the in-sample's
 - Standard deviation or risk are less than the first portfolio's
 - Sharpe Ratio is then, greater too indicating that the out-of-sample one provides better risk-adjusted returns .
- 



Simple Risk Parity II - Comparison

IN-SAMPLE

```
risk_in_SimpleRP = 0.0066  
ret_in_SimpleRP = -5.3865e-05  
SharpeR_in_SimpleRP = -0.0082
```

OUT-OF-SAMPLE

```
risk_out_SimpleRP = 0.0053  
ret_out_SimpleRP = 6.5612e-04  
SharpeR_out_SimpleRP = 0.1228
```



The data reveals that in this case out-of-sample portfolio heavily outperforms the other due to:

- Expected return has turned from negative values to positive ones
 - Standard deviations or risks is lower
 - Sharpe Ratio shows that the risk-adjusted return for the second portfolio is clearly higher and thus, the latter is preferred.
- 



Risk Parity II - Comparison

IN-SAMPLE

$\text{risk_in_RP} = 0.0079$
 $\text{ret_in_RP} = -3.8838\text{e-}05$
 $\text{SharpeR_in_RP} = -0.0049$

OUT-OF-SAMPLE

$\text{risk_out_RP} = 0.0072$
 $\text{ret_out_RP} = 9.6750\text{e-}04$
 $\text{SharpeR_out_RP} = 0.1335$

The data reveals that in this case out-of-sample portfolio also outperforms the other in a great extent since:

- Expected return is well above the other, from negative to positive terms
 - Standard deviations or risks is also lower in the second
 - Sharpe Ratio shows that the risk-adjusted return for the second portfolio is also, evidently higher and thus, the latter is preferred.
- 
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04 Conclusion

Target p-value		Markowitz		Equal Weight		Simple Risk Parity		Risk Parity		
In sample	0.01	In sample	Out of sample	In sample	Out of sample	In sample	Out of sample	In sample	Out of sample	
Out of sample	0.0057									
30-70		Risk	0.0086	0.0117	0.0060	0.0065	0.0057	0.0062	0.0061	0.0081
		Return	0.0027	-0.00026	0.000589	0.00040	0.000151	0.0007	0.00013	0.0006
		Sharpe Ratio	0.3126	-0.0225	0.0987	0.0625	0.0267	0.0599	0.0213	0.0746
50-50		Risk	0.0109	0.0102	0.0070	0.0056	0.0064	0.0053	0.0079	0.0072
		Return	0.0020	-0.00005	0.00035	0.00066	-0.00005	0.0007	-0.0000	0.001
		Sharpe Ratio	0.1794	-0.0054	0.0506	0.1175	-0.0079	0.1228	-0.0048	0.1335

30-70 MV vs RP vs HRP (EW as benchmark)	Beta	Treynor's Ratio	Alpha	Max_DD	Sortino_Ratio
In-Sample MV	0,089909543	0,005114492	0,000418497	0	26,58552865
In-Sample RP	1,231602875	0,000373368	-0,000106501	-1,444829991	12,72156773
In-Sample HRP	0,304721403	0,001509056	0,000319718	-1,375241554	5,179233046
Out-of-Sample MV	0,125093149	0,002214357	0,00024235	65535	2,554494274
Out-of-Sample RP	1,102161093	0,000251325	-2,82987E-05	-2,852268309	13,85357096
Out-of-Sample HRP	0,192336	0,001440193	0,000223724	-7,129318109	9,824694161

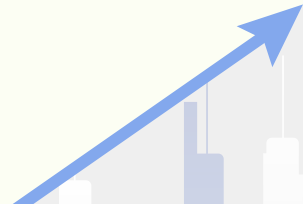
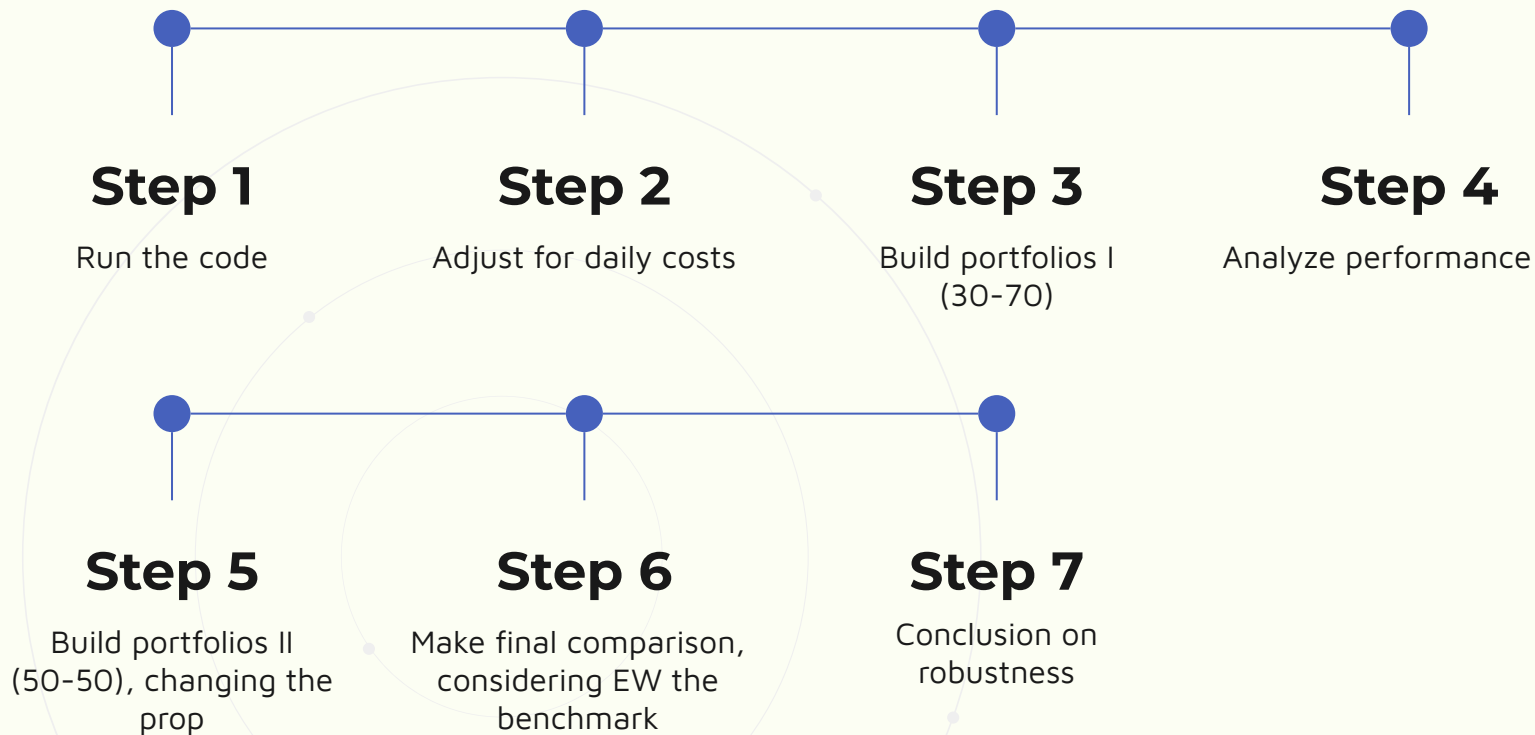
Could be wrong, correct results on the code

50-50 MV vs RP vs HRP (EW as benchmark)	Beta	Treynor's Ratio	Alpha	Max_DD	Sortino_Ratio
In-Sample MV	0,13519	0.0016057	0.00019299	0	13.826
In-Sample RP	1.2687	0.0001759	-5.9953e-05	-10.116	8.5373
In-Sample HRP	0.29855	0.00074745	0.00015653	-5.4794	3.3483
Out-of-Sample MV	0.11081	0.0048179	0.00047472	-Inf	4.8518
Out-of-Sample RP	1.0353	0.0005157	-1.8821e-05	-0.84729	21.964
Out-of-Sample HRP	0.1404	0.0038025	0.00045892	-1.5573	67.167

Could be wrong, correct results on the code



What have we done?





Thanks!

Do you have any questions?

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